

“Report on: e-values for multiplicity control in multiverse analysis”

Summary

The paper presents a comparison of multiverse analysis under different approaches (universal e-values, soft-rank e-values, and p-to-e calibration). This is performed through simulation studies and a real data application. Overall, I find the paper very interesting and well-written, but I have some comments that may help improve the presentation of the paper. Specific comments are provided below.

Specific comments

Abstract

- It may be worth clarifying that the comparison of the different approaches is performed through a simulation study.
- It may also be worth specifying the calibrator that produces the best performance.

Section 1

- Some references (or at least a representative one) regarding the applications mentioned in the first paragraph would be welcomed.
- Similarly, in the second paragraph, some reference on multiverse analysis would be particularly welcomed, as this is not necessarily a mainstream topic.
- The critique of Specification Curve Analysis (SCA) feels a bit heavy-handed. Perhaps the authors could consider briefly acknowledging situations where SCA is valid or widely accepted before detailing its pitfalls.

Section 2

- Page 5. The subpopulation-encoding notation (g_{ik}) relies on a specific weighted sum-to-zero constraint. It would be useful for the authors to include a brief, intuitive explanation of how this constraint behaves in practice, as the current mathematical definition is quite dense.
- Page 5. Could you please clarify if the definition of treatment effect in terms of the derivative of $F \circ \mathbb{E}$ is mainstream or a specialized definition for your paper? Marginal effects are often defined in terms of the derivative of $\mathbb{E}$ (Ai and Norton 2003) instead, which accounts for the scale induced by the link function. Of course, this does not affect linear regression, but it induces a difference for other GLMs.
- Page 6. The authors state control covariate effects (η_z) are always assumed non-zero, in contrast to SCA's subset approach. Could you please briefly justify why including all controls is universally preferable here, as over-controlling (e.g., inducing collider bias) can sometimes be a risk in observational multiverse data. Would a DAG or other tool help understand when this strategy is safe?

Section 3

- Some references on e-values would be welcomed at the beginning of this section.
- Page 8. Should the prior (4) be conditional on σ^2 ? It may also be worth clarifying that the Zellner's prior allows for a unit information prior interpretation in the context of linear regression, while for other GLMs it may be needed to use a variance of the type $W^T M(\beta) Z$, in line with the Fisher information matrix associated with other GLMs. The choice $W^T W$ looks quite reasonable to me, specially to avoid diluting the main message of the paper, but a brief discussion may be welcomed.
- Page 9. It may be worth mentioning that model misspecification in GLMs may also involve misspecifying the link (e.g. in binary regression).]
- I wonder if the simulation study would be better placed as a separate section.
- The simulation study utilizes 10 covariates drawn from a multivariate Gaussian with a uniform pairwise covariance of 0.5. Could the authors discuss this specific covariance structure. Is this moderate-to-high level of collinearity representative of typical Social Science datasets?
- In Table 1, the Type I error is exactly 0 for many sample sizes at both the 1% and 5% levels across multiple e-variables. Could the authors dedicate a sentence to discussing whether these e-variables are too conservative in finite samples, and how that impacts the false negative rate. Or is this an effect of the number of Monte Carlo iterations?

- Please, report (general) computing times to give the reader an idea of the different computational costs, which are previously mentioned.

Section 4

- I wonder if the title of this section could be made more informative, in line with the real data application.
- Were there any missing values in the data set? If so, how were they handled?
- The authors reports different conclusions to the original studies. A slightly more extensive discussion on why that is the case would be welcomed.

References

Ai, C., and E. C. Norton. 2003. “Interaction Terms in Logit and Probit Models.” *Economics Letters* 80 (1): 123–29.